

FORGE STAR

Africa's sovereign satellite data network

Business model · Build strategy · Stages · Capital · Potential

MYFORGE PTY LTD · FORGE ECOSYSTEM

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01 · EXECUTIVE SUMMARY

FORGE STAR in one page

FORGE STAR is Africa's sovereign satellite data network: compliant, locally owned LEO infrastructure that connects the assets, vessels and people that terrestrial networks will never reach.

FORGE STAR is a South African company in the FORGE ecosystem (MYFORGE PTY LTD). It is building a neutral, sovereign low-Earth-orbit (LEO) communications network for Africa and the Global South: a third option for governments and operators that do not want to depend on Starlink (US) or Guowang/Qianfan (China).

The wedge is **satellite IoT, maritime AIS and sovereign data relay**, not consumer broadband. That is fundable now, serves customers that intermittent coverage already satisfies, and avoids a head-on fight with a ~10,000-satellite incumbent.

Dimension	FORGE STAR position
Customers	Maritime/security agencies, mining, agriculture, water, conservation, FORGE ecosystem companies; later SA government connectivity, mobile operators and other African states
Offer	Sovereign, locally owned satellite data and connectivity anywhere in Africa, data landed in South Africa
Build approach	Open-source software across the stack + heypcb.ai for boards + Claude Code for integration; satellite bus bought from SA partners
Path	Ground first (revenue before orbit) → hosted payload → 6–12 satellites → 24–48 satellite operational network
Capital to operational network	~$50–105M gross; ~$40–75M equity after revenue, grants and prepayments
Bootstrappable	First ~2 years: Stage 0 and a hosted payload in orbit for ~$0.3–1M
Most likely success outcome	Stage 3 operator worth ~$200–500M; $1B+ if African governments co-finance continental scale

Why now

- **Starlink is blocked in South Africa** by ICASA's 30% historically disadvantaged ownership rule, which SpaceX will not meet. FORGE STAR can comply.
- **Chinese constellations are far behind target** (a few hundred satellites against ~28,000 planned) and carry alignment risk.
- **OneWeb is enterprise-only in SA**; Amazon Leo is still preparing entry; the EU's IRIS² has no satellites in orbit.
- **Tooling has collapsed early costs:** open-source flight, ground and NTN software plus AI-assisted design make a demonstrator a tens-of-thousands problem, not a millions problem.

02 · THE COMPANY

What FORGE STAR is, and how the plan was sharpened

The original concept was an Exosat-style mega-constellation: ~$300k satellites at 420–450 km, ITAR/EAR-free, selling IoT, Direct-to-Cell and broadband. Stress-testing it produced seven refinements that make it buildable from South Africa.

Issue in the original plan	Refinement
Direct-to-Cell on a $300k small satellite does not close: phones need very large antennas	Lead with 3GPP Rel-17 NB-IoT NTN and store-and-forward messaging; D2C later through a partner, not our hardware
A PocketQube demonstrator proves software, not the business	Fly the real payload: hosted payload first, then a 6U–12U demonstrator
Building our own bus wastes engineering	Buy the bus from SA suppliers (Dragonfly Aerospace, CubeSpace); own the payload, network, ground and operations
30% ownership rule treated only as a moat	It also binds us: structure the cap table for it from day one, and watch equity-equivalent reforms that could let Starlink in
Mega-filing through ICASA is unrealistic	File a modest NGSO network (24–100 satellites) in IoT-appropriate bands; achievable ITU milestones
Conflicting launch and revenue figures	Budget $80–250k per 6U–12U slot via broker; revenue built from unit math (Section 09)
Too many products at once	One wedge first; earn the right to D2C, broadband and edge compute

The three payload functions

- **NB-IoT NTN and store-and-forward messaging** for mining, agriculture, water and utility metering, logistics and conservation.
- **Satellite AIS/VDES** for maritime domain awareness across South Africa's very large EEZ: Navy, fisheries enforcement, ports.
- **Sovereign data relay** for government sensors (border, environment, disaster), landed in South Africa to support POPIA and data-sovereignty needs.

The FORGE ecosystem advantage

FORGE venture	What FORGE STAR gives it
FORGE Freight	Container and truck tracking across African trade corridors
Maritime USV venture	Over-the-horizon command link and AIS
FORGE VOCAL	Satellite messaging/USSD fallback for feature-phone users in no-coverage zones
ForgePay	Offline-tolerant payment settlement over satellite messaging
FORGE EYE	Sovereign sensor and data feed into the decision platform
Autonomy studio / robotics	Remote telemetry for field robots and vehicles

03 · TARGET MARKET

Who we serve and what we offer them

Beachhead customers (Stages 0–2)

Customers who can use intermittent coverage and pay early.

Segment	Example buyers	What we offer	How they pay
Maritime and security	SA Navy, DFFE fisheries, SAMSA, Transnet ports, shipping insurers	Satellite AIS vessel tracking, illegal-fishing detection	Annual data subscription or government contract
Mining and energy	Mining houses, Eskom, renewable IPPs, pipeline operators	Remote monitoring of dams, pumps, generators, grid sensors, vehicles	Per device per month + platform fee
Agriculture and water	Commercial farms, co-ops, water boards	Soil, water-level, irrigation and livestock sensors beyond cell coverage	Per device per month
Conservation	SANParks, private reserves, anti-poaching units	Animal collars, fence sensors, ranger messaging	Per device or per reserve
FORGE ecosystem	Freight, USV, VOCAL, ForgePay, EYE	Tracking, vessel links, offline messaging and payments	Internal contracts; first references
Ground station services	Foreign satellite operators	Southern-hemisphere downlink passes	Per pass or per month

Expansion customers (Stage 3+)

Segment	Example buyers	What we offer	How they pay
SA government connectivity	DCDT / SA Connect, provinces, municipalities, Education, Health	Managed rural connectivity for schools, clinics, hotspots; sovereign data relay; disaster comms	Per site per month, multi-year
Mobile operators	MTN, Vodacom, Telkom, Cell C, Rain	Wholesale NB-IoT NTN capacity and coverage extension; continental distribution	Wholesale per device/MB + revenue share
Logistics corridors	Rail, trucking, cold-chain across SADC	Cross-border cargo and fleet tracking without roaming gaps	Per asset per month
Other African governments	SADC and wider African states	Constellation-in-a-box: their flag and filing, our satellites and operations	Upfront co-financing + revenue share

04 · BUSINESS MODEL

How FORGE STAR makes money

We charge for IoT and maritime data per device and per contract, sell sovereign capacity to governments under multi-year agreements, and use customer co-financing so we do not carry the full constellation cost ourselves.

Revenue line	Customer	Pricing
IoT connectivity (NB-IoT NTN + store-and-forward)	Mines, agriculture, utilities, logistics, conservation, FORGE Freight	$2–5 per device per month, volume tiers
Maritime data (AIS/VDES)	Navy, fisheries, ports, insurers	Annual data subscription
Sovereign capacity	SA government departments	Multi-year capacity contract, partly prepaid
Ground station services	Third-party satellite operators	Per pass or per month
Constellation-in-a-box (later)	Other African states	Upfront fees + revenue share on co-financed satellites
Mobile operator wholesale (later)	MTN, Vodacom and others	Wholesale IoT/messaging capacity

Why the model works

- **Captive early demand** from the FORGE ecosystem gives reference customers before the first satellite.
- **Customers fund the build:** sovereign buyers prepay or co-finance capacity, moving capex off our balance sheet.
- **Local-ownership compliance** lets FORGE STAR operate where Starlink currently will not.
- **IoT pricing benchmarks:** new LEO IoT entrants price from roughly $1 to €8 per device per month; legacy satellite IoT sits at $40–70. FORGE STAR competes on sovereignty and local service at competitive prices.

Could South Africa have free internet, paid by government?

Free to citizens, paid by government, is a sound model, but not with our own satellites alone early on. Serving 30 million users at 10 GB/month needs ~1 Tbps average (~3 Tbps peak) of capacity, which means thousands of broadband satellites: Starlink-scale capital.

Approach	Who gets connected	Feasibility
Satellite backhaul to government sites (schools, clinics, police stations, hotspots)	~10–20k rural sites, free WiFi around each	**Feasible.** Leased capacity first, own broadband layer later
Hybrid national plan: fibre/towers in cities, FORGE STAR for the last 10–15%	Rural and remote communities	**Best model.** Aligns with SA Connect
Everyone, everywhere, on our satellites only	All ~63 million people	**Not feasible** without Starlink-scale capital

The sellable version: FORGE STAR as government's sovereign operator for rural connectivity, paid per connected site per month, starting on leased capacity from Stage 0 and migrating to our own satellites later.

05 · BUILD STRATEGY

What we build, what we buy, and the full component stack

- **We buy:** the satellite bus (structure, power, attitude control) from SA partners such as Dragonfly Aerospace and CubeSpace; antennas and solar panels.
- **We build:** the payload (NB-IoT NTN radio + AIS receiver), the network, the ground segment and data platform, and sovereign operations. This is our IP.
- **We accelerate:** with open source across the stack, heypcb.ai for boards, and Claude Code for integration, testing and analysis.

Layer	Components	Job
Flight software	NASA F´ (primary), NASA cFS (alt.), libcsp, Zephyr / FreeRTOS	Onboard command, telemetry, scheduling, fault handling, subsystem messaging, real-time OS
Reference hardware	LibreCube, BIRDS CubeSat designs	Proven board and bus starting points
Mechanical / electronics	FreeCAD, KiCad	Structure, enclosures, board design files
RF and payload	GNU Radio, SoapySDR, gr-satellites, SatDump, AIS-catcher / gr-ais	Signal processing, SDR abstraction, decoding, AIS payload
IoT radio and core	OpenAirInterface NTN, Open5GS / free5GC	NB-IoT NTN stack and 5G core
Ground segment	SatNOGS, Gpredict + Hamlib, Yamcs or OpenC3 COSMOS, NASA Open MCT, TinyGS	Ground network, antenna control, mission control, dashboards, LoRa backup
Flight dynamics	Orekit, NASA GMAT, Skyfield	Orbit determination, manoeuvre planning, pass calculation
Constellation design	OpenSN	Coverage, revisit and routing emulation
Customer platform	ThingsBoard, Apache Kafka, PostgreSQL + TimescaleDB	Device management, data pipeline, SA-hosted storage

Where heypcb.ai fits

heypcb.ai is AI-assisted circuit-board design: schematics, routing, Gerbers, BOMs and KiCad projects, with a Circuit World library of 8,800+ open-hardware boards to fork. **It builds:** payload processor boards, NB-IoT/AIS radio carrier boards, payload-to-bus interface boards, ground-station and test electronics. **It does not handle:** structure, thermal design, high-performance RF and antennas, radiation qualification, or bus integration.

In-house PCB pipeline (open-source alternative to heypcb.ai)

`hardware/` builds boards from code with open-source tools only. SKiDL describes the circuit using KiCad's libraries. KiCad's Python API places the parts and applies the rules. Freerouting autoroutes. KiCad checks the design and exports Gerbers, drill files, pick-and-place data and a BOM. The first board is the **payload interface board**: an STM32L432 running the flight core, CAN to the bus, a radio power switch, current and temperature telemetry. It passes the electrical rule check with no errors and is placed with no blocking DRC issues. Autorouting is the next step.

Proven open CubeSat boards (LibreCube, OreSat, PyCubed, BIRDS) replace heypcb's library of boards to fork. ngspice and openEMS cover circuit and antenna simulation. **We can do the board work in house.** As with heypcb, it doesn't cover radiation qualification, thermal design, RF tuning or bus integration.

Flight software, radio link and customer API

- **Flight core (`flight/libforge`).** Portable C99 with no dynamic memory: the store-and-forward buffer, frame encoding and Reed–Solomon error correction. Its output is byte-identical to the Python ground code, which the shared test vectors prove.
- **F´ component (`flight/fprime`).** `StoreForward` wraps the core with commands, telemetry and events. It's built and unit-tested in NASA F´ v3.5.1, and the ground decodes its downlink.
- **Radio link layer.** CCSDS concatenated coding: Reed–Solomon (255,223) in the dual basis, plus the K=7 rate-1/2 convolutional code with a soft-decision Viterbi decoder, the CCSDS randomiser and a sync marker. It decodes at about 2.5 dB Eb/N0, against about 6 dB for Reed–Solomon alone. A software BPSK modem handles Doppler, timing and phase.
- **Command security.** Commands are ChaCha20-encrypted and HMAC-SHA256-authenticated, with replay protection, and checked on board before anything executes.
- **Customer API (`ground/api`).** Per-customer keys, devices, messages, delivery SLA and monthly usage, with data hosted in SA.

Can we physically build the satellite?

Yes, with the bus bought in. The four things that decide whether it is actually usable are: **(1) permission to transmit** (ICASA licence + ITU filing; without it a satellite is scrap in orbit), **(2) qualification** (vibration, thermal vacuum, radiation), **(3) launch** (achievable but external, rideshare via broker, dates slip), and **(4) something on the ground to talk to** (NTN devices and enough coverage for the use case).

06 · COVERAGE

How many satellites South Africa, SADC and Africa need

LEO satellites cannot be parked over South Africa: they circle Earth roughly every 95 minutes. Covering SA continuously means covering its whole latitude band worldwide, so SA, SADC and Africa need similar numbers. **Revisit time matters more than region size.**

Service level	User experience	South Africa	SADC	All of Africa
Store-and-forward IoT	Data delivered within a few hours	2–6	4–8	6–12
Frequent IoT	Data within 20–40 minutes	16–24	24–36	30–48
Continuous narrowband	Real-time messaging; always one satellite overhead	50–90	60–110	80–150
Broadband	Internet-grade speeds	Set by capacity demand, not coverage		

Original first-order estimates. Checked below by simulation.

Simulation check (November 2026)

`sim/coverage_sim.py` simulates Walker constellations at 550 km over 13 sites from Cape Town to Tunis, including the SA EEZ and Marion Island. It includes Earth's rotation and J2 drift, and runs over three days. The elevation mask comes from the link budget (below). The worst site in each region is shown.

Constellation	Simple patch (47° mask): SA typical / longest wait	Isoflux antenna (16° mask): SA typical / longest wait	Isoflux: Africa longest wait	Gateway passes/day
1 sat, SSO rideshare	36 h / 58 h	~10 h / 13 h	14 h	3
6 sats, 45°	3.1 h / 9.7 h	~61 min / 5.8 h	10 h	27
12 sats, 45°	1.6 h / 7.2 h	~29 min / 5.0 h	5.4 h	56
24 sats, 45°	48 min / 4.8 h	~12 min / 1.6 h	3.9 h	113
48 sats, 45°	23 min / 3.1 h	~6 min / 66 min	1.9 h	221
80 sats, 50°	16 min / 2.1 h	~4 min / 12 min	62 min	320
120–150 sats, 50°	10–14 min / ~80 min	0–2 min / 0–3 min	18–19 min	480–600

The mask is where a device can both hear the satellite and reach it. With a simple patch antenna that is about 47° (the downlink sets it). With the shaped isoflux antenna and about 8 W of RF power it is about 16°. **The antenna decides the service**: 48 satellites with a patch are worse than 24 with the isoflux antenna.

What changes:

- **With the isoflux antenna, the original numbers hold up.** 24 satellites give SA a typical wait of about 12 minutes.
- **The longest wait is worse than assumed.** Promising "every 20–40 minutes, guaranteed" needs **~48–80 satellites for SA** and **~80–120 for Africa**, with the isoflux antenna.

- **Sell typical latency, not worst case**, until Stage 3. Customer SLAs should be written as "95% of messages within X". The platform's `latency_sla` view measures exactly this.
- **A 45° inclination beats sun-synchronous (SSO)** for Africa. A first rideshare will probably go to SSO, which is fine for a demo but not for service. Mixing the two in one constellation leaves long gaps.
- **For continuous narrowband, 80–150 is right**, and only with the isoflux antenna or better.

Link budget check

`forgestar/linkbudget.py` uses illustrative 6U–12U parameters. Replace them with datasheet values. Antenna gain is now tied to beam shape: a ±30° beam has about 9.7 dBi, not the 14 dBi first assumed (corrected November 2026 after an independent review).

Link	Closes (3 dB margin) from	Implication
NB-IoT uplink (23 dBm device → 9.7 dBi sat patch, ±30° beam)	~29° elevation	Not the limit
NB-IoT downlink (2 W into the same patch)	~47° elevation	**Sets the mask, and is the payload design driver.** Devices must hear the downlink before they can send. Fix it with a shaped (isoflux) or steerable antenna, more RF power, deeper repetition, or a narrower downlink carrier. Budget RF engineering for this
AIS receive (VHF monopole)	horizon	AIS is limited by message collisions, not link budget
Feeder downlink (X-band → 3.7 m SA dish)	~10°	Closes at the 10° gateway mask with 3 dB margin

Antenna trade study

`sim/antenna_trade.py` compares payload antennas. The usable mask is the higher of uplink and downlink. Service is shown for the worst SA site.

Option	Usable mask	24 sats: typical / longest wait	Stage 3 (48): typical / longest wait	DC power	Est. cost
Patch ±30°, 2 W (baseline)	47°	48 min / 4.8 h	28 min / 1.9 h	12 W	~$15k
Patch + deep coverage mode (software)	34°	30 min / 4.2 h	16 min / 72 min	12 W	~$15k
Patch, 10 W amplifier	32°	26 min / 3.9 h	15 min / 56 min	45 W	~$40k
Isoflux to 30°, 6 dBi, 5 W	23°	19 min / 3.1 h	11 min / 38 min	25 W	~$45k
Isoflux to 20°, 5 dBi, 8 W	16°	11 min / 1.6 h	7 min / 12 min	38 W	~$55k
Isoflux, 5 W, at 450 km	17°	17 min / 2.5 h	10 min / 22 min	25 W	~$45k
Isoflux, 5 W + deep coverage mode	7°	7 min / 32 min	3 min / 5 min	25 W	~$45k
Isoflux, 2 W, compact own protocol	10°	8 min / 50 min	4 min / 8 min	12 W	~$35k
4×4 phased array, 4 W	20°	16 min / 2.2 h	10 min / 22 min	30 W	~$150k

Recommendation: fly the demonstrator with the simple patch plus deep coverage mode, which is a software change. Its coverage is poor by design. Build the Stage 2–3 payload around the isoflux antenna with about 8 W, which needs about 38 W from the bus during passes: a 12U, and a question for the bus partner. If the bus cannot supply it, the fallbacks are 5 W with deep coverage mode, a 450 km orbit, or the compact protocol. Keep the phased array for Stage 4.

Capacity per pass

`sim/capacity_study.py` asks how many messages one satellite can carry. One NB-IoT carrier takes about 6 messages a second. Most early devices sit in southern Africa, inside one footprint, so the satellites over that region set the limit.

Fleet and payload	Messages per device per day at 40k / 110k / 330k devices
Stage 3, NB-IoT, 1 carrier	11 / 4 / 1.4
Stage 3, deep coverage mode	2.8 / 1.0 / 0.3
Stage 3, compact own protocol	32 / 12 / 3.9
Stage 4 (150 sats), NB-IoT, 1 carrier	32 / 12 / 3.9
Stage 4, phased array with 4 carriers	130 / 47 / 16

- **At scale, capacity is a tighter limit than coverage.** Stage 3 carries about 1.4 messages per device per day at 330,000 devices. Hourly reporting for every device does not fit.
- **Sell message plans by tier.** A few readings a day fits Stage 3. Hourly plans need Stage 4, more carriers or the compact protocol.
- **Deep coverage mode trades capacity for reach.** Use it for hard-to-reach devices only.

Simplified model: circular orbits with J2. Checked against SGP4 (`sim/sgp4_crosscheck.py`): orbit planes agree to within 3 km over 3 days, and the remaining difference is a timing offset equal to 1–3 km of altitude. Firm up with Orekit, then OpenSN, before external use. All antenna figures are illustrative.

Delivery time: satellites vs gateways

`sim/latency_sim.py` follows each message end to end. It waits for a satellite overhead, rides on board, then lands at the first gateway pass. Devices re-send on each pass until the ground acknowledges.

Setup (95% of messages delivered within…)	Johannesburg	Maputo corridor	Lusaka	Dar es Salaam	Nairobi	Lagos
24 sats, 16° mask, 4 gateways	77 min	57 min	74 min	2.7 h	3.7 h	3.8 h
36 sats, 16° mask, 4 gateways	44 min	71 min	12 min	44 min	1.8 h	1.9 h
48 sats, 16° mask, 4 gateways	42 min	53 min	9 min	50 min	1.9 h	1.8 h

Gateways: Hartebeesthoek, Cape Town, Walvis Bay, Mauritius. A 16° mask needs the shaped (isoflux) antenna.

- **Gateways are the cheap lever** for sites far from SA. They can be leased passes or partner stations relaying data to the SA platform, so data still lands in SA.
- **The 60-minute SLA is reachable across SADC** with about 36–48 satellites, the isoflux antenna and 4 gateways.
- **Equatorial sites (Nairobi, Lagos) stay around 2 hours.** They need a low-inclination plane or equatorial gateways before an SLA is offered there.

- **Devices must re-send until acknowledged.** Handing each message to a single satellite can make waits worse, because that satellite may be heading away from every gateway. This is now built: ACK frames, gateway duplicate removal, satellites dropping acked copies, devices stopping re-sends. In a live platform run on 24 satellites with SA gateways only, 95% of messages landed within about **40 minutes**, against 85 minutes with the single-copy policy.

Closing the equatorial gap

`sim/equatorial_study.py` (isoflux antenna, re-send until acked) gives 95% delivery time in minutes:

Option	Jhb	Agulhas	Maputo	Lusaka	Dar	Nairobi	Lagos	Extra capex vs 36 sats
36 @45°, 4 gateways	44	10	71	12	44	110	116	—
+ Malindi + Lagos gateways	44	10	69	10	44	96	60	~$0.7M
+ 6 sats @10° + Malindi + Lagos	44	10	69	10	21	10	25	~$5.2M
42 @45° + 6 @10° + Malindi + Lagos	8	6	8	26	11	9	11	~$9.7M

Stage 3 design: 42 satellites at 45° plus 6 in a 10° plane (48 in total) and 6 gateways (Hartebeesthoek, Cape Town, Walvis Bay, Mauritius, Malindi, Lagos). Every customer site is within 26 minutes for 95% of messages. A Maputo gateway does not help: the Maputo corridor is limited by the wait for a satellite overhead, which the extra 45° satellites fix.

Key takeaway: a 24–48 satellite constellation gives the whole continent useful IoT and maritime data on average, which is why the Africa-wide pitch works early. Tight worst-case revisit needs ~48 (SA) to ~80 (Africa) satellites. Real-time continuous coverage needs ~80–150 (Iridium-class).

07 · STAGES

What is possible at each point, and what our build strategy covers

Stage	Satellites	Timing	What works	Capital
0: Ground first	0	Months 0–12	Ground stations, IoT platform, resold capacity, rural sites on leased capacity, licences lodged	$1–3M
1: Demonstrator	1–2	Months 12–30	Store-and-forward IoT, AIS snapshots, full chain proven, government demo	$5–12M
2: Early service	6–12	Years 2.5–3.5	Africa-wide IoT in 1–3 h; useful maritime picture	$15–30M
3: Operational network	24–48	Years 3.5–5	IoT in ~30 min across Africa; ITU filing secured; sovereign contract; MNO wholesale	$30–60M + prepayments
4: Continuous narrowband	80–150	Years 5–7	Real-time messaging/IoT; emergency messaging to NTN phones; other states	$100–250M, co-financed
5: Broadband layer	Hundreds+	Years 7+	Own capacity for national rural connectivity programme	$0.5–2B+, gov./DFI

Schedule check (November 2026)

`readiness/tracker.py` chains the 30 outside tasks in `readiness/readiness.yaml` by their dependencies. These are spectrum, ownership, funding, lab, orbit and customers.

Plan	Stage 2 (6–12 sats) ready	From start
Everything in sequence	Jan 2032	~61 months
Seed raised alongside the HDG-anchor search	Dec 2031	~60 months
+ own demonstrator built alongside the hosted payload	Dec 2030	~48 months
+ Series A on the hosted-payload result, tighter builds	Jul 2030	~43 months

The stage table's "years 2.5–3.5" for early service is **only reachable on the fast track**, and even then it lands at the late end, about 3.6 years. The biggest lever is to **build our own 6U–12U demonstrator in parallel with the hosted payload, not after it**. That saves about a year. It needs a seed round large enough to fund both. The second lever is to **raise the Series A on the hosted-payload result**. Both should become plan decisions.

Durations are planning estimates. Update `readiness/readiness.yaml` as quotes and ICASA timelines firm up.

Build-strategy coverage by stage

Stg	Open source carries	heypcb.ai builds	Not covered (we fill)	Tech / total
0	SatNOGS, Gpredict, Yamcs/COSMOS, Open MCT, ThingsBoard, Kafka, TimescaleDB, OAI NTN lab, OpenSN, Orekit	Ground-station and lab test boards	Leased capacity, ICASA applications, 30% structure, pilots, sites	~90% / ~50%

Stg	Open source carries	heypcb.ai builds	Not covered (we fill)	Tech / total
1	F´/cFS, libcsp, Zephyr, GNU Radio, SatDump, AIS-catcher, OAI NTN, LibreCube/BIRDS, FreeCAD/KiCad	Payload processor, radio carriers, bus interface, payload power	Bus, antennas/RF tuning, qualification, spectrum licence, launch, insurance	~60% / ~30%
2	Same flight/payload software reused; Orekit/GMAT fleet planning; OpenSN; Yamcs automation	Rev-2 boards for batch build, fixtures	Batch manufacture/QA, launches, SADC ground stations, landing rights, sales	~55% / ~25%
3	Full stack at fleet scale; Kafka/TimescaleDB; OAI + Open5GS for MNO integration	Production variants, gateway electronics	Production line, ITU bring-into-use, security accreditation, 24/7 NOC	~50% / ~20%
4	Software reusable; OpenSN routing; OAI NTN (heavy in-house work)	Digital/control boards only	Inter-satellite links, larger satellites, scale manufacturing, multi-country spectrum	~35% / ~15%
5	Ground, ops and platform software; network simulation	Support electronics only	Phased arrays, Ka/Ku payloads, terminals, teleports, broadband spectrum	~15–20% / ~5–10%

Open source plus heypcb.ai carries Stages 0–2 to a large degree: a working satellite network for tens of millions of dollars, not hundreds. From Stage 3 the value shifts to what we own: payload IP, manufacturing, spectrum, government contracts and operations.

08 · DISTANCE TO LAUNCH

How much of the path to full launch our approach covers

"Full launch" means the 24–48 satellite operational network. Contribution = component share × coverage.

Component	Share	Open source + heypcb.ai	+ Claude Code / Claude assist	Total covered	Contribution
Software (flight, ground, network, platform)	20%	80%	~70% of remainder	~94%	18.8%
Simulation and mission design	5%	90%	~80% of remainder	~98%	4.9%
Electronics (boards)	10%	55%	~25% of remainder	~66%	6.6%
Payload RF and antennas	10%	20%	~25% of remainder	~40%	4.0%
Satellite bus	10%	0%	~10% (interfaces)	~10%	1.0%
Testing and qualification	10%	0%	~20% (automation, analysis)	~20%	2.0%
Spectrum and licensing	10%	0%	~25% (filings, interference analysis)	~25%	2.5%
Launch	5%	0%	~5% (paperwork)	~5%	0.25%
Capital raising and sales	20%	0%	~15% (decks, models, tenders)	~15%	3.0%
Total	**100%**	**~28%**			**~43%**

Open source + heypcb.ai covers ~28% of the whole process to launch. **Adding Claude Code takes it to ~43%**, with the biggest gains in software, payload RF analysis, spectrum filings and sales material. The remaining ~57% is physical, regulatory or relational: buying hardware, lab qualification, ICASA/ITU decisions, launch slots, and winning customers and investors.

Claude accelerates rather than replaces: engineers still review and sign off, especially on flight software and RF. The practical effect is that a lean team of 4–6 can carry Stage 0–1 workloads that would normally need 10–15 people.

09 · CAPITAL

What it costs, and how far we can bootstrap

Stage	Satellites	Stage capital	Cumulative	Lean team
0: Ground first	0	$1–3M	$1–3M	4–6
1: Demonstrator	1–2	$5–12M	$6–15M	8–12
2: Early service	6–12	$15–30M	$21–45M	15–25
3: Operational network	24–48	$30–60M	$51–105M	40–70
4: Continuous narrowband	80–150	$100–250M	$150–355M	100+
5: Broadband layer	Hundreds+	$0.5–2B+	$0.65–2.4B+	Hundreds

Gross figures. Revenue, grants and prepayments reduce the equity need to reach Stage 3 to ~$40–75M. Stages 4–5 are funded mainly by government and development-finance co-financing.

Financial model check (November 2026)

`model/financial_model.py` builds years 1–6 from `model/assumptions.yaml`, which sets out device counts, prices, team, satellite unit costs, launch, ground stations and funding.

Check	Strategy	Model (base)	Comment
Stage 3 revenue (36 sats)	$25–45M	~$37M	Consistent, **if** 330k devices and one SA government contract land
Equity to Stage 3	$40–75M	~$50M	Consistent
Gross capital to Stage 3	$51–105M	~$32–49M (high to low case)	The strategy's range is conservative. Keep it as the budget, since it includes buffers the model doesn't
Stage 0 spend	$1–3M	~$0.7M lean	The two numbers describe different plans. Lean bootstrap ≈ $0.5–1M, funded Stage 0 ≈ $1–3M
Cash low point		Year 2	Low case runs short by ~$0.9M in year 2. Close the seed round before the hosted payload is paid for

Monthly cash. Even the base case goes about $0.5M negative for two months at the start of year 2. That's when hosted-payload and first-satellite payments come in before the seed money lands. **Close the seed round by the end of year 1**, or phase the payload deposits. The low case needs a larger seed round, or an earlier Series A.

Ownership. The cap table (`model/assumptions.yaml` → `ownership`) keeps HDG ownership above 50% through Series B, as long as about a quarter to a third of each round comes from HDG-qualifying investors (DFIs, BEE funds) and half the ESOP goes to HDG employees. If the rounds come only from foreign VCs, the ratio falls fast. Plan an HDG anchor investor in every round, or a broad-based trust. Founder HDG flags are placeholders. Confirm each holder's status and the current rule with counsel.

Stress tests (`model/assumptions.yaml` scenarios, 48 satellites, 4 leased gateways):

Scenario	Year-6 revenue	Year-6 EBITDA	Note
Base	$37.5M	$12.6M	Seed closed by month 12; no month below zero cash
IoT at ~$1/device	$31.6M	$9.2M	Still inside the Stage 3 range
No SA government capacity contract	$19.5M	$2.2M	Below the Stage 3 range: **the biggest commercial risk**

IoT price pressure. New entrants now price at about $1 per device per month (Myriota from $0.99; Sateliot targeting about €1). That's below our $2–5 assumption. At about $1, year-6 revenue falls from about $37.5M to about $31.6M, which is still in range. The reason: IoT is only about a quarter of Stage 3 revenue, and the **SA government sovereign-capacity contract is the line the plan really depends on**. Price IoT on sovereignty and local support, not on undercutting, and win the government anchor early.

The model's biggest sensitivity is **IoT device adoption**. Revenue from 330k devices at about $3 per month only comes if a mobile operator or a big mining or logistics customer signs. A 40% revenue shortfall still reaches Stage 3, but only with the Series B on time.

Bootstrapping: revenue first, payload in orbit second, equity third

Phase	Funding source	Spend	Team	What we have at the end
Bootstrap (years 0–1.5)	Revenue, grants, founder equity	~$0.2–0.5M	4–6	Ground network, IoT platform, reseller customers, licence filed, lab-proven payload
Lean demo (years 1.5–2.5)	Grants, prepayments, small angel round	~$0.5–3M	8–12	Our payload proven in orbit (hosted ~$200–500k, or own 6U ~$1.5–3M)
First raise (year 2.5+)	Series A $15–30M	Stage 2	15–25	Africa-wide IoT constellation
Growth (years 3.5+)	Series B + government prepayments	Stage 3	40–70	Operational network, sovereign contract

Stage 0 revenue streams

- **IoT reseller:** resell an existing satellite IoT network under the FORGE STAR brand with our platform on top; move customers to our satellites later.
- **Ground station services:** sell southern-hemisphere passes to foreign operators.
- **Managed rural connectivity:** run government and mining sites on leased capacity; thin margins, but it builds the sovereign relationship.
- **FORGE ecosystem contracts:** Freight and the maritime venture as first paying users.

Non-dilutive sources to target: TIA, DSI, SANSA partnerships (facilities, ground stations), NEF and IDC, DTIC manufacturing and R&D incentives. Verify current calls.

What bootstrapping covers: ~80–100% of the first launch (hosted payload), ~40–60% of our own demonstrator, and ~10% of the work (~1% of the capital) to full launch. It is the stretch that removes the most risk, so the remaining 90% raises on strong terms.

10 · COMPETITIVE LANDSCAPE

FORGE STAR against the incumbents

Competitor	Scale (late 2026)	Sovereignty posture	FORGE STAR's angle
Starlink	10.3M paid subscribers (Mar 2026, S-1), ~12M by mid-2026; ~10,800 working satellites; ~300k African subscribers (unverified)	US-controlled; licence-blocked in SA by ownership rules	Do not compete on broadband; win on sovereignty, compliance and local ownership
Guowang / Qianfan	A few hundred satellites combined, far behind targets	Chinese state-aligned	Displace in the non-aligned segment
Eutelsat / OneWeb, IRIS²	OneWeb ~650 satellites, enterprise-only in SA; IRIS² services from ~2030	European sovereign, not African	Leaves African sovereign and mass IoT open
Amazon Leo	Hundreds of satellites, enterprise beta, preparing SA entry	US / AWS-adjacent	Future competitor; not serving African sovereign demand yet
Exosat	Pre-revenue; ~11,000-satellite ITU filing; LOIs claimed $400M+	Singapore-based; Asia/Middle East focus	Closest comparable; FORGE STAR is Africa-first
IoT specialists (Kinéis, Myriota, Sateliot)	Niche constellations	Mostly Western	Compete on sovereignty and local service, priced competitively

Figures are approximate as of the research done for this plan and should be re-verified before external use.

Structural advantages

- **Credible non-alignment** for African and Global South buyers.
- **Real local industry:** CubeSat heritage (ZACUBE, MDASat), CubeSpace (ADCS to 350+ missions), Dragonfly, the MeerKAT/SKA RF talent pool, an automotive supply base.
- **Continental distribution** through SA-headquartered operators (MTN, Vodacom).
- **Co-financing model** means no need to raise billions alone.

The lean founding team

- Founder/CEO for government and business development
- RF/SDR payload engineer
- Software and ground-segment engineer (full open-source stack)
- Embedded/electronics engineer (heypcb.ai)
- Part-time spectrum and regulatory advisor
- Sales lead for mining, agriculture and conservation

11 · POTENTIAL

What FORGE STAR can become

Stage	Annual revenue	Illustrative valuation	Driver
0–1: Ground + demonstrator	R5–30M	$5–30M	Team, licences, payload proven in orbit
2: Early service (6–12)	$3–10M	$50–150M	Africa-wide IoT, first contracts
3: Operational (24–48)	$25–45M	$200–500M	Sovereign contract, MNO wholesale
4: Continuous (80–150)	$100–300M	$1–3B	Continental coverage, other states as customers
5: Broadband layer	$500M+	$3–10B+	National connectivity programmes

Stage 3 revenue build (illustrative)

Line	Assumption	Annual revenue
IoT connectivity	300k devices × $3/month	~$11M
AIS / maritime data	3–5 subscribers	~$3–6M
Sovereign capacity	1 SA government deal	~$10–25M
Ground station services	Third-party passes	~$1–2M
Total	~40 satellites, ~$15–25M fleet capex	**~$25–45M**

Comparables: Sateliot and Kinéis (early satellite IoT), Iridium (continuous narrowband, multi-billion), Eutelsat/OneWeb and Telesat (sovereign-anchored operators). Most likely successful outcome is a $200–500M Stage 3 company and an attractive acquisition target; $1B+ depends on government co-financing. Illustrative ranges, not a formal valuation or financial advice.

12 · RISKS AND NEXT STEPS

Where it breaks, and the next 90 days

Risk	Why it matters	Mitigation
Spectrum (ICASA / ITU)	Binding constraint; no licence means no service. ITU milestones truncate filings if deployment lags	Small, specific filing; specialist counsel; early ICASA engagement; backstop administration option
Ownership rules change	Equity-equivalent reforms could let Starlink in	Do not build the pitch on Starlink staying out; sell sovereignty and IoT, not broadband
Qualification and RF	Payload must survive launch and space and meet link budgets	Hosted payload first; partner bus; qualification budget; RF hires
Capital	Thin SA seed market; Stage 2 needs $15–30M	Bootstrap to proof; grants; raise Series A from strength; blended international capital
Government sales cycles	LOIs are not contracts; procurement is slow	Revenue-generating Stage 0 services; paid pilots; ecosystem anchor demand
Launch slippage	Rideshare dates move	Diversified brokers and providers; schedule buffers

Next 90 days

- **Spectrum:** brief specialist counsel; request an ICASA pre-application meeting on NGSO IoT/AIS bands.
- **Bus partners:** approach Dragonfly Aerospace and CubeSpace for partnership and 6U–12U quotes; explore hosted-payload slots.
- **Anchor demand:** internal LOIs from FORGE Freight and the maritime venture; outreach to one mining group and one fisheries or Navy contact.
- **Ownership:** design the 30% ownership structure with a BEE specialist.
- **Funding:** TIA/DSI grant application for the NB-IoT NTN payload lab prototype (OpenAirInterface on SDR).
- **Ground first:** stand up the first SatNOGS-based station and the ThingsBoard/Kafka platform; sign an IoT reseller agreement.
- **Regulatory watch:** confirm current status of Starlink and equity-equivalent rules in South Africa.

Bottom line: FORGE STAR is buildable from South Africa if it runs as a ground and services company before it becomes a satellite company. The tools make the first two years cheap; spectrum, qualification, capital and government contracts set the schedule. Sequenced that way, the plan has a real path to a $200–500M sovereign operator, with continental upside.

Not financial advice.